

classify_to_functional_type.R

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classify_to_functional_type
<i>Classify Community Data to Functional Types</i>

Description

Classifies taxa in a long-format community data frame to functional-type (FT) labels using a pre-computed FT classification table. Aggregates pollen proportions by all non-‘taxon’/‘value’ identifier columns plus functional type. Produces the same output column structure as ‘classify_taxonomic_resolution()’ so it can be used as a drop-in replacement downstream.

Usage

```
classify_to_functional_type(  
  data_source,  
  data_functional_type_classification,  
  flag_verbose = TRUE  
)
```

Arguments

data_source	A data frame containing community data with columns ‘taxon’, ‘dataset_name’, ‘age’, and ‘value’. Other identifier columns such as ‘sample_name’ are preserved.
data_functional_type_classification	A data frame mapping taxa to functional types. Must contain columns ‘taxon_name’ (character) and ‘functional_type’ (integer). Typically produced by ‘cluster_functional_types()’ and loaded via ‘get_functional_type_classification()’.
flag_verbose	Logical. If ‘TRUE’ (default), informational progress messages are printed to the console via ‘cli’. Note: the warning issued when taxa are absent from ‘data_ft_classification’ is always emitted regardless of this flag.

Details

Steps performed:

1. Validate arguments.
2. Left-join 'data_source' to 'data_functional_type_classification' on their taxon names.
3. Drop unmatched taxa (NA functional type) with a warning.
4. Create 'taxon' labels "FT_functional_type".
5. Aggregate 'value' by original identifier columns and 'taxon'.
6. Full-join back to an identifier/taxon cross-reference to preserve true negatives.

Value

A data frame with the same column names as 'data_source'. The 'taxon' column is replaced by functional-type labels of the form "FT_1", "FT_2", etc. 'value' is aggregated (summed) by all original identifier columns and 'taxon'. All identifier combinations present after FT classification are preserved (true negatives kept via a cross-reference join). Taxa not found in 'data_functional_type_classification' are dropped with a 'cli::cli_warn()' message.

See Also

[classify_taxonomic_resolution()], [get_functional_type_classification()], [cluster_functional_types()]

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